

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim:- Simulate linear convolution and circular convolution on discrete time signals.	
Open handed problem -2	Date: 05-08-2026	Enrolment No:92400133131

Aim :- Simulate linear convolution and circular convolution on discrete time signals.

Theory:

Linear convolution and circular convolution are mathematical operations used to combine two signals to obtain a third signal. They are widely used in various applications, such as signal processing, image processing, and audio processing.

Linear convolution calculates the sum of element-wise products of two signals, considering the full range of valid indices. It is typically used for finite-length signals and can produce an output signal that is longer than the input signals.

Circular convolution, on the other hand, calculates the sum of element-wise products of two signals, considering a periodic extension of the input signals. It is commonly used for periodic or infinite-length signals and produces an output signal with the same length as the input signals.

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```

CODE:- from pydub import AudioSegment
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import convolve

audio = AudioSegment.from_mp3(r"D:\sem-5\dsip\Shararat Dhurandhar 320 Kbps.mp3")

audio = audio.set_channels(1)
samples = np.array(audio.get_array_of_samples()).astype(np.float32)

kernel = np.array([1, 0, 1, 0, 1], dtype=np.float32)

convoluted = convolve(samples, kernel, mode="same")

convoluted = convoluted / np.max(np.abs(convoluted))
convoluted_int16 = (convoluted * 32767).astype(np.int16)

convoluted_audio = AudioSegment(
    convoluted_int16.tobytes(),
    frame_rate=audio.frame_rate,
    sample_width=2,
    channels=1
)

convoluted_audio.export("output_convoluted.wav", format="wav")

print("Convoluted audio saved as 'output_convoluted.wav'.")

samples_norm = samples / np.max(np.abs(samples))
convoluted_norm = convoluted / np.max(np.abs(convoluted))

plt.figure(figsize=(12, 6))

plt.subplot(2, 1, 1)
plt.plot(samples_norm, color="blue")
plt.title("Original Audio Signal")

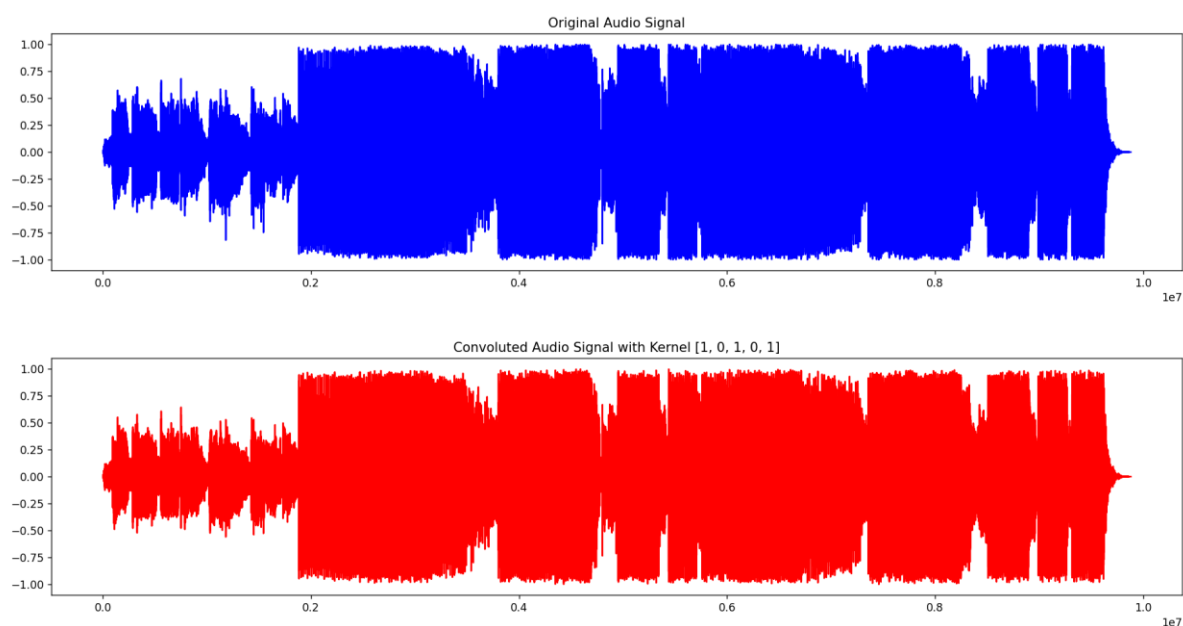
```

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```
plt.subplot(2, 1, 2)
plt.plot(convoluted_norm, color="red")
plt.title("Convolutud Audio Signal with Kernel [1, 0, 1, 0, 1]")

plt.tight_layout()
plt.show()
```

OUTPUT :-



1. mode="same"

Role

- Keeps the output the same size as the original audio.
- Most useful for filtering.

Graph:

- Same width as the original graph.

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- Easier to compare with the original signal.

2. mode="full".

Role

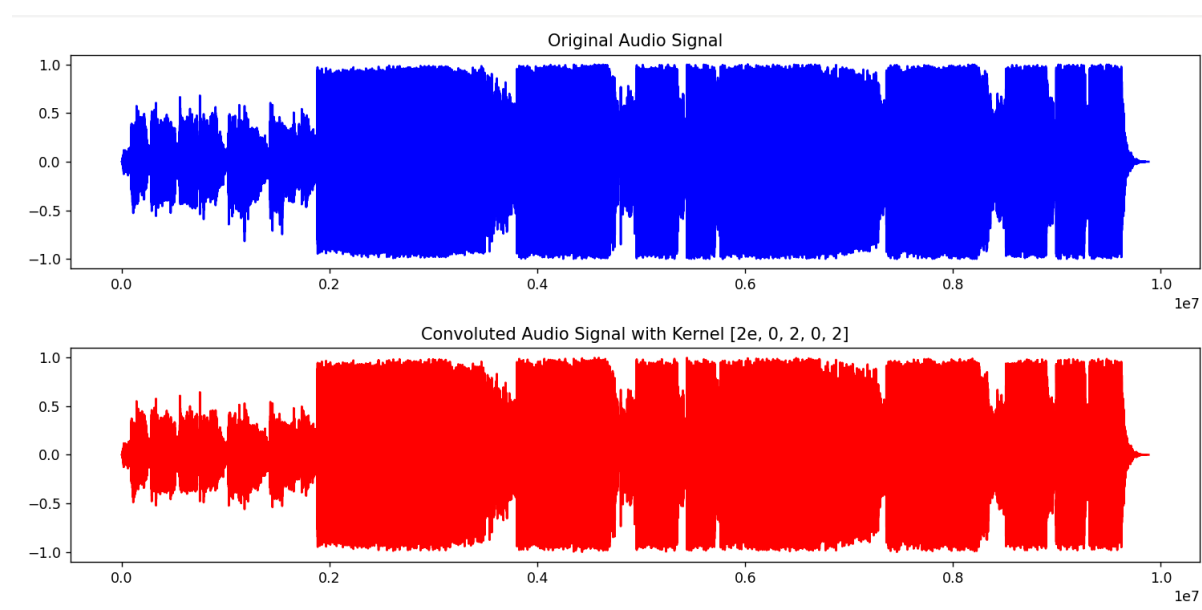
- Uses every overlap between the signal and kernel.
- Includes start and end effects.

Role

- Uses every overlap between the signal and kernel.
- Includes start and end effects.

Graph:

- Longest waveform.
- More samples than the original.



3. mode="valid"

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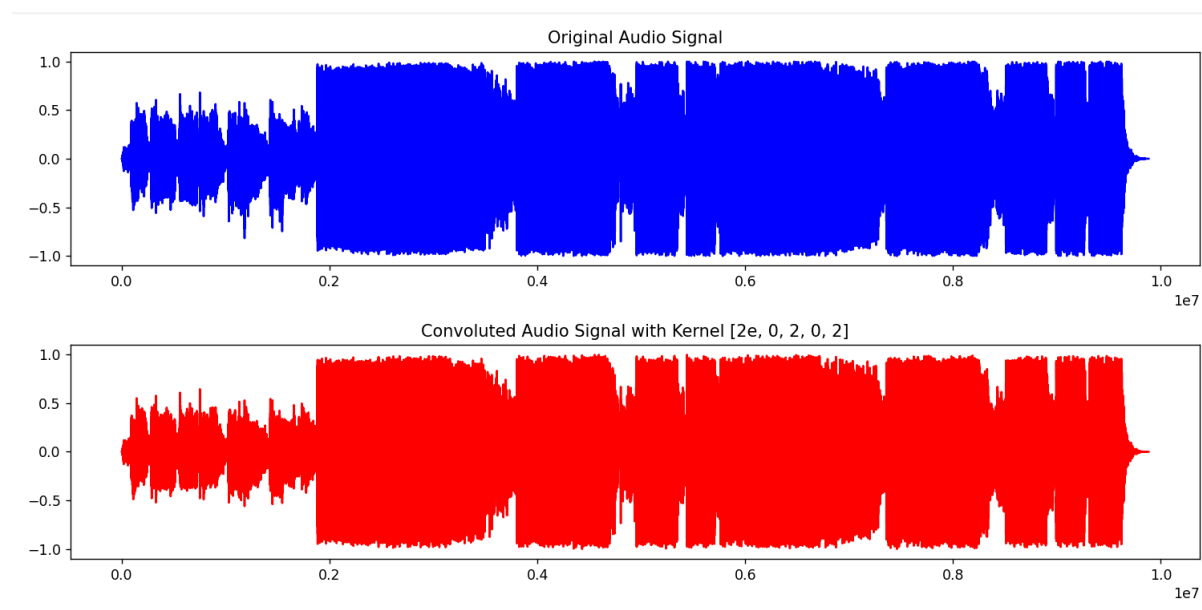
Role

- Uses only positions where the kernel is completely inside the signal.
- Removes boundary effects.

Graph:

- Shortest waveform.
- Beginning and ending parts are removed.

Output :-



```
kernel = np.array([0, 0, 0, 0, 0], dtype=np.float32)
```



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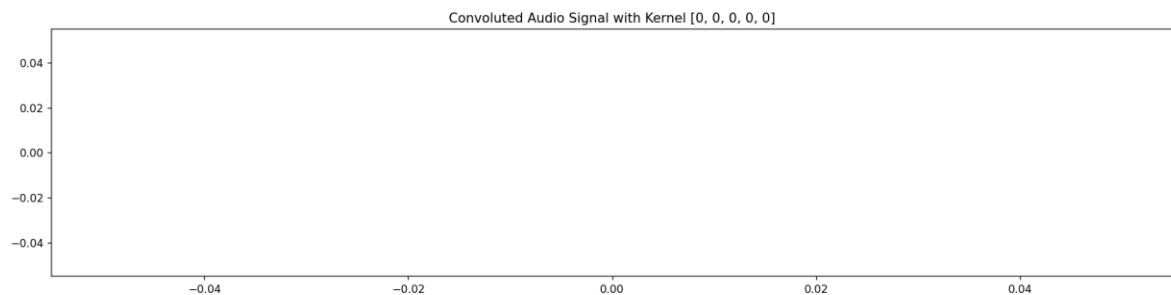
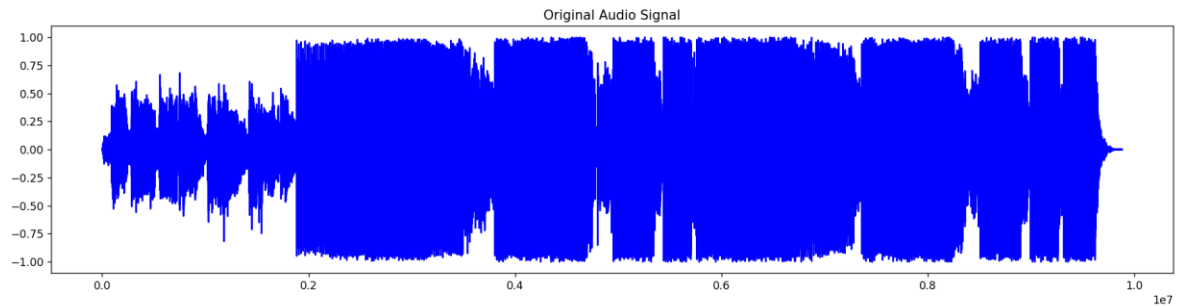
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discrete time signals.**

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```
kernel = np.array([1, 1, 1, 1, 1], dtype=np.float32)
```



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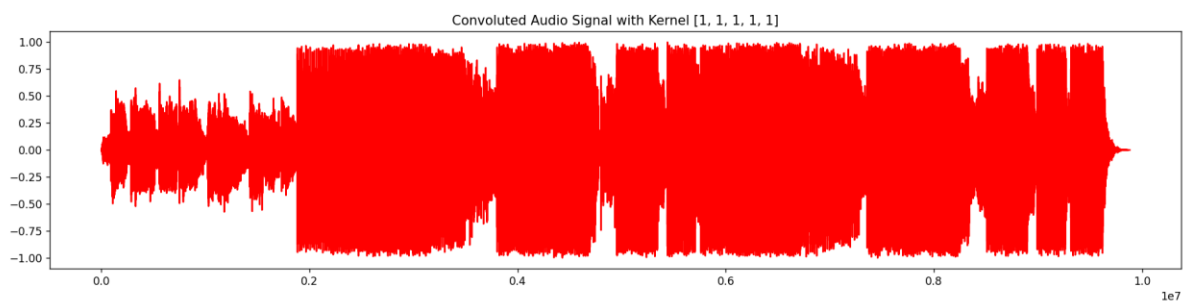
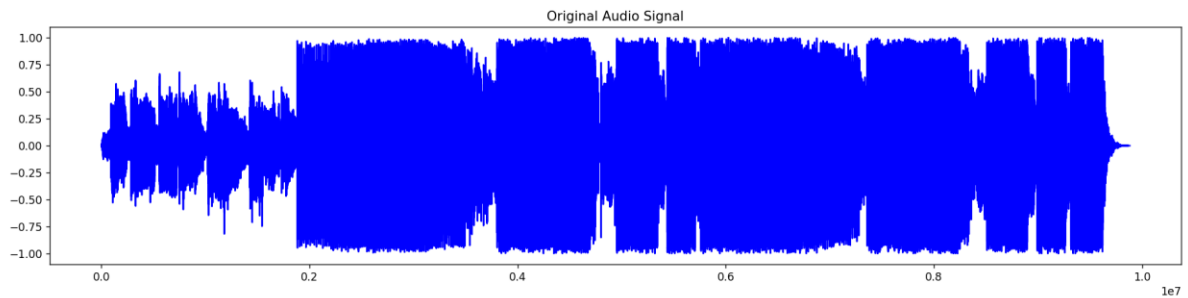
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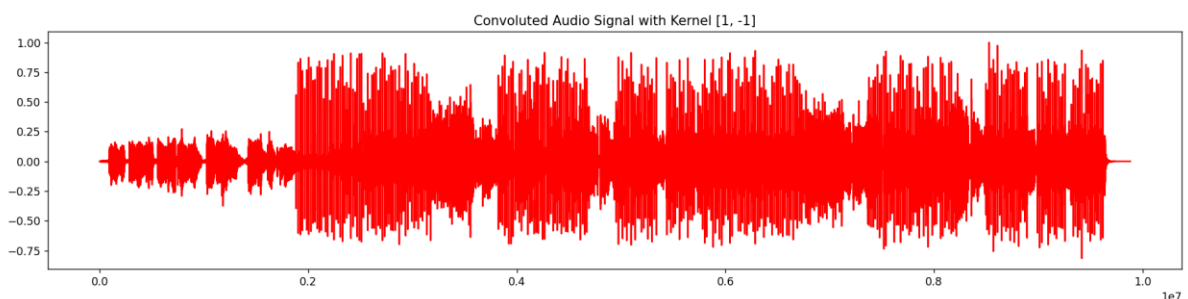
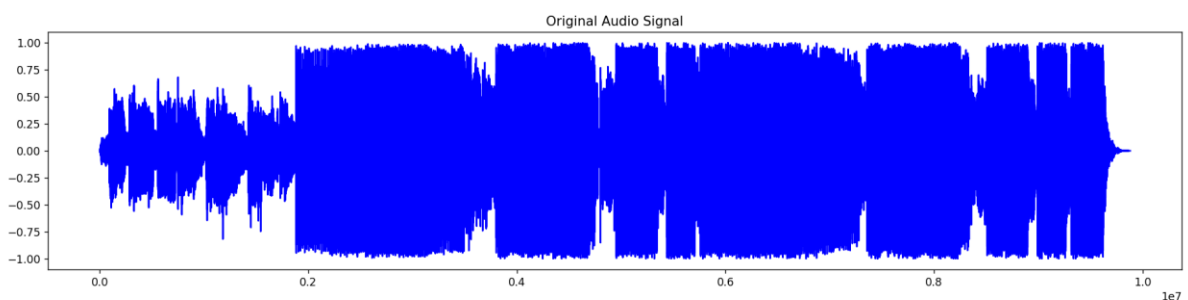
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```
kernel = np.array([1, -1], dtype=np.float32)
```



```
kernel = np.array([0.2, 0.2, 0.2, 0.2, 0.2], dtype=np.float32)
```



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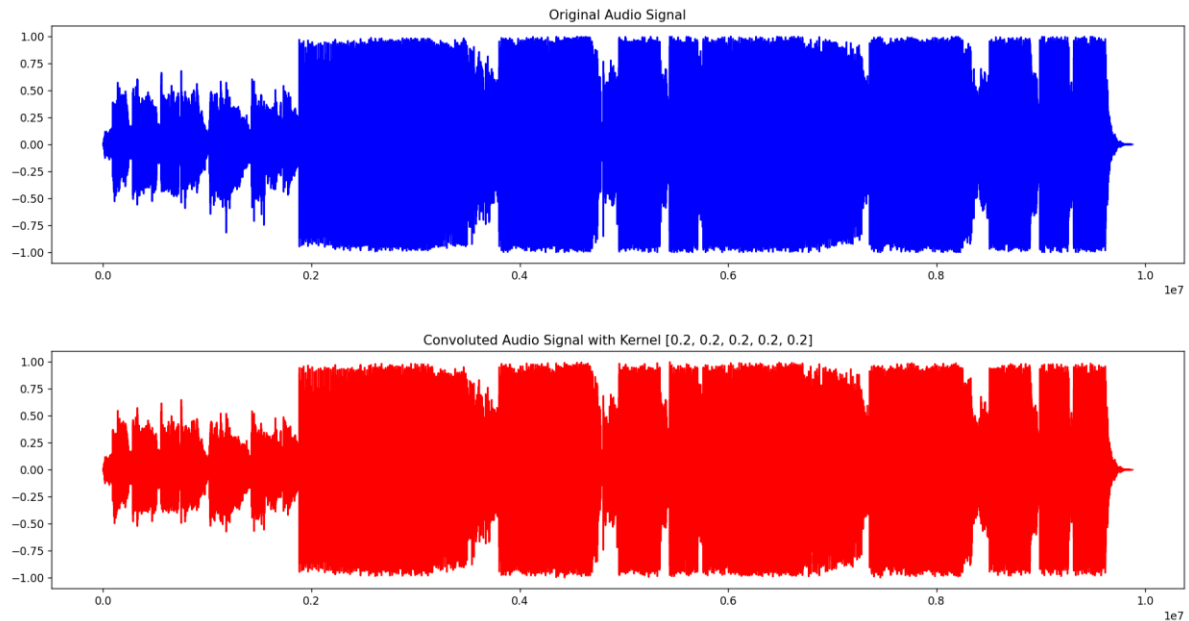
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```
kernel = np.array([1, 0, -1], dtype=np.float32)
```



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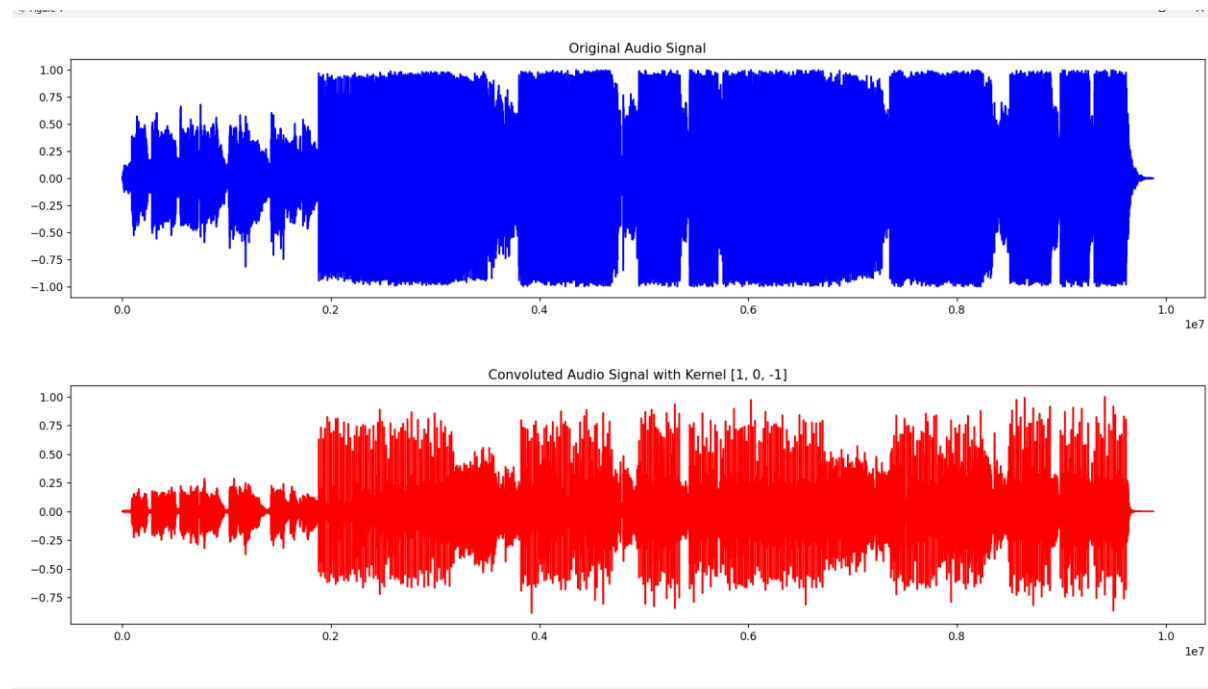
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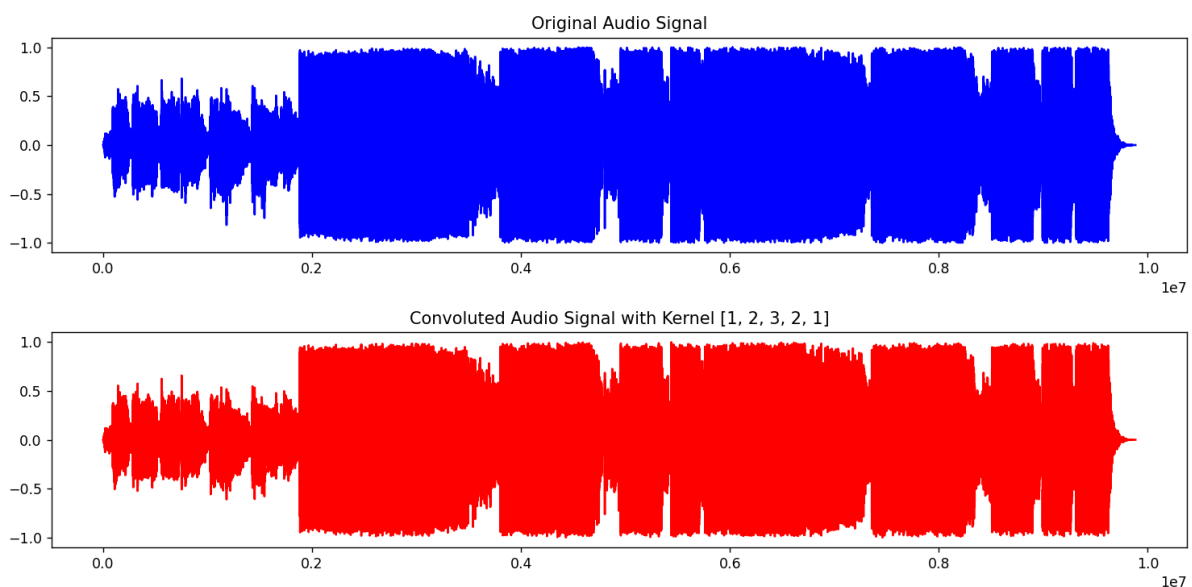
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```
kernel = np.array([1, 2, 3, 2, 1], dtype=np.float32)
```





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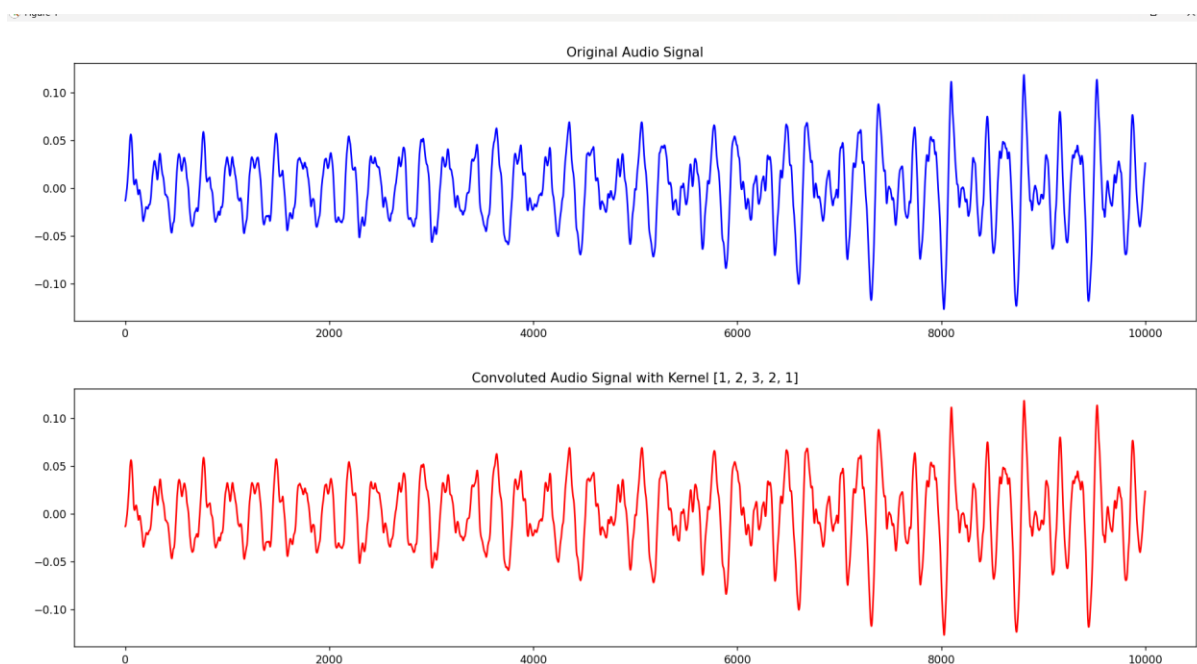
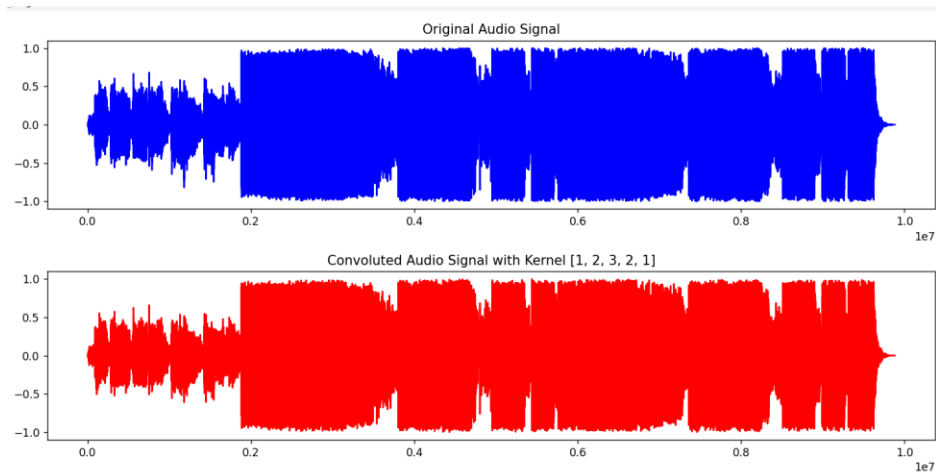
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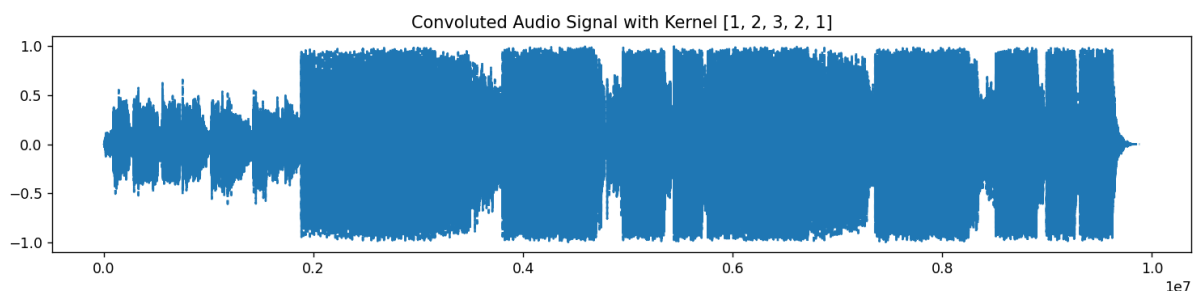
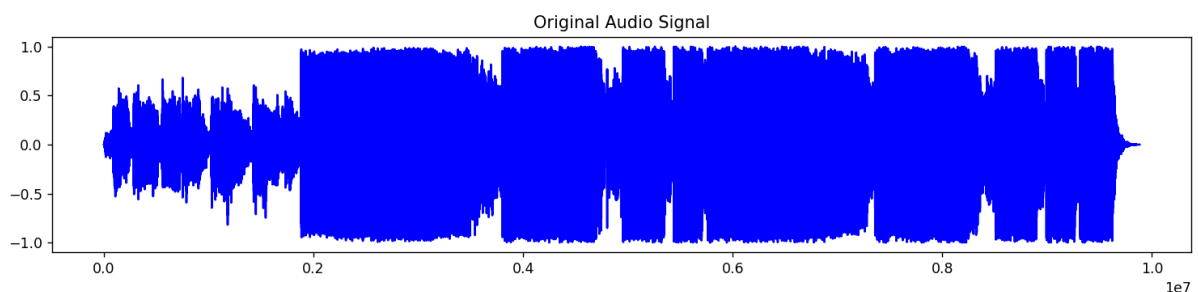
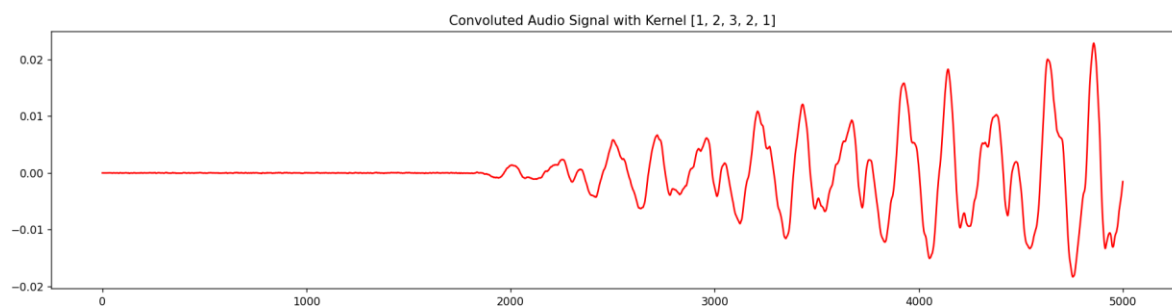
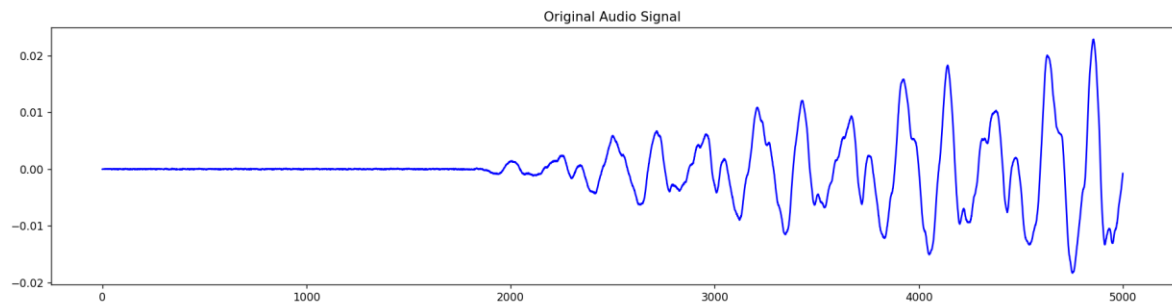
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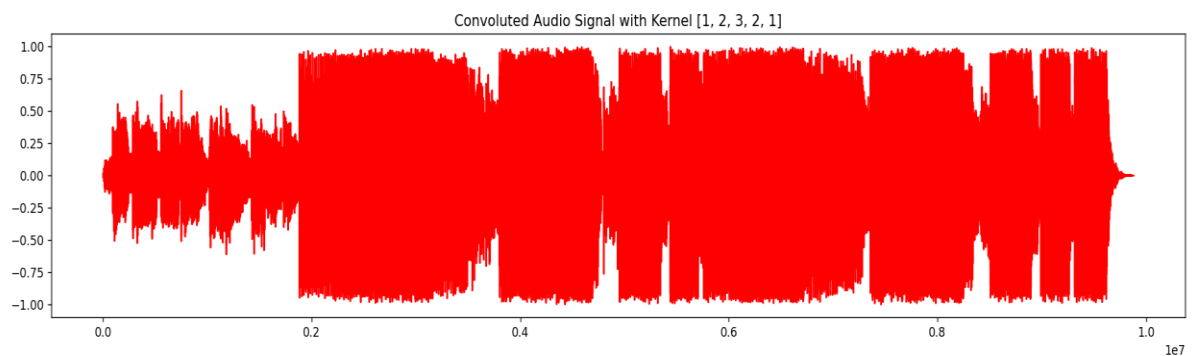
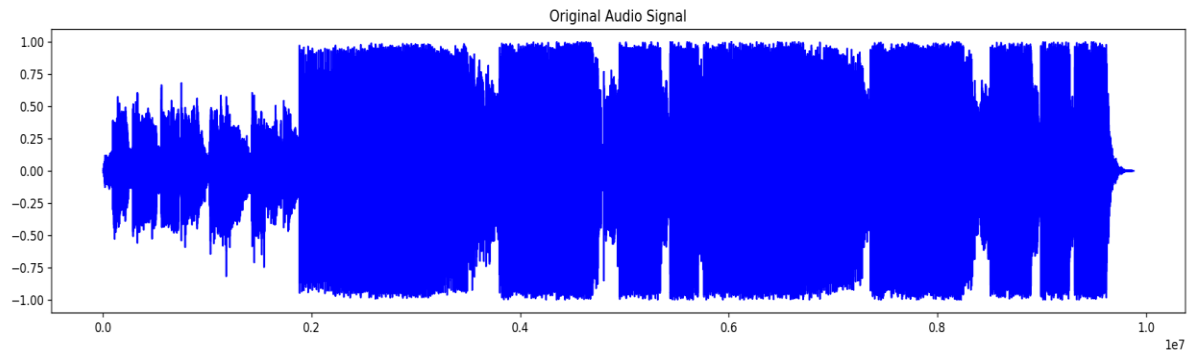
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```
kernel = np.array([1], dtype=np.float32)
```



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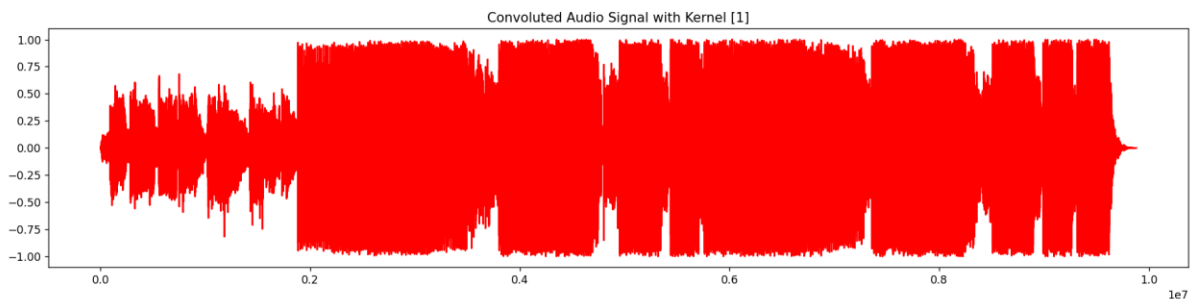
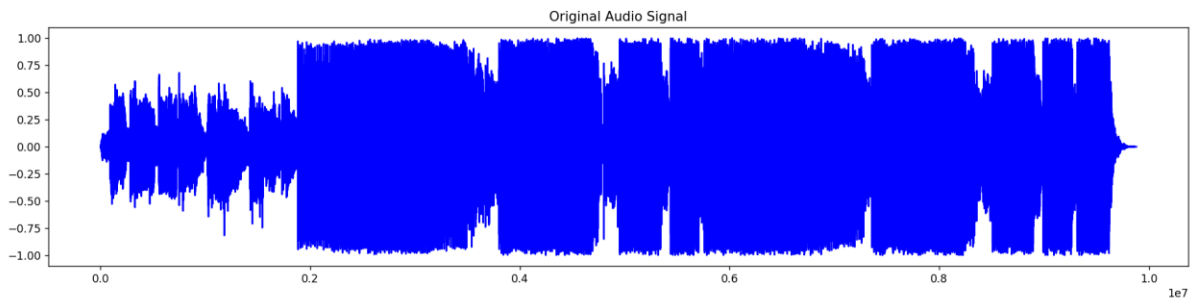
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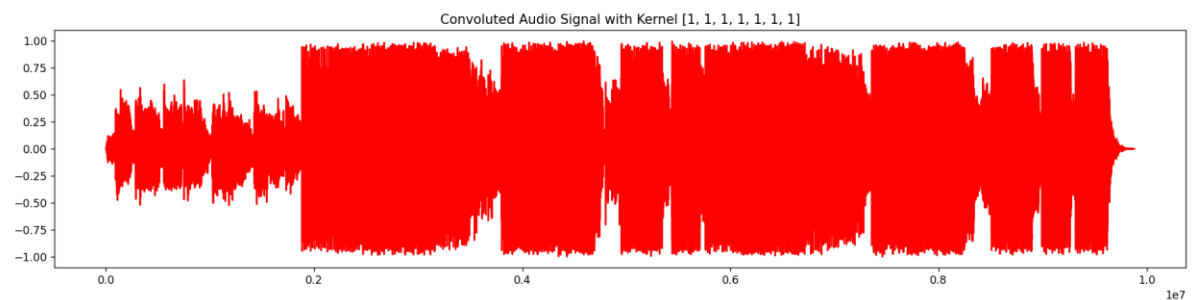
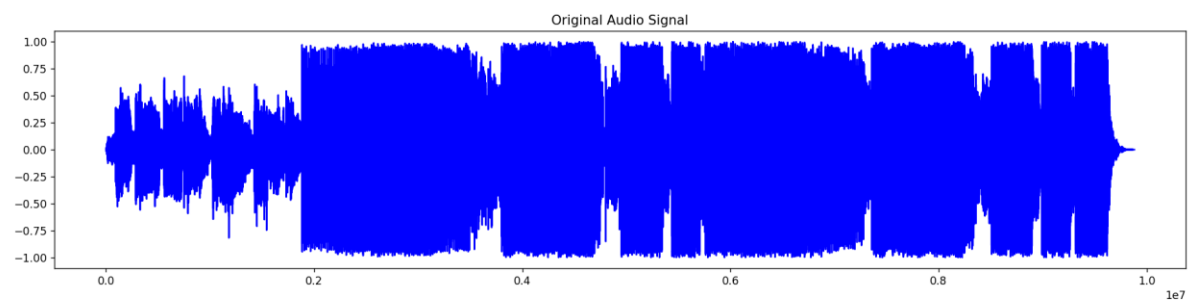
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```
kernel = np.array([1, 1, 1, 1, 1, 1, 1], dtype=np.float32)
```



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1. Change the Kernel

```
kernel = np.array([1, 0, 1, 0, 1], dtype=np.float32)
```

```
kernel = np.array([1, 1, 1, 1, 1], dtype=np.float32)
```

```
kernel = np.array([1, -1], dtype=np.float32)
```

```
kernel = np.array([0.2, 0.2, 0.2, 0.2, 0.2], dtype=np.float32)
```

```
kernel = np.array([1, 0, -1], dtype=np.float32)
```

```
kernel = np.array([1, 2, 3, 2, 1], dtype=np.float32)
```

1. mode="full"

Definition:

Returns the **complete convolution** of the two signals. It includes all overlapping positions between the input signal and the kernel.

Formula:

- Output length = $N + M - 1$

where:

- N = length of the input signal
- M = length of the kernel

Code :- `convoluted = convolve(samples, kernel, mode="full")`

Role:

- Computes the entire convolution result.
- Includes the beginning and ending edge effects.

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- Best for studying convolution in DSP.

Graph:

- Longest output graph.

2. mode="same"

Definition:

Returns the convolution result with the **same length as the input signal**.

Formula:

- Output length = **N**

convoluted = convolve(samples, kernel, mode="same")

Role:

- Keeps the output size equal to the input size.
- Commonly used for signal filtering and image processing.
- Makes it easy to compare the original and filtered signals.

Graph:

- Same length as the original signal.

3. mode="valid"

Definition:

Returns only the convolution values where the kernel **completely overlaps** the input signal.

Formula:

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- Output length = $N - M + 1$

convoluted = convolve(samples, kernel, mode="valid")

Role:

- Removes boundary (edge) effects.
- Uses only fully valid overlaps.
- Often used when only the most accurate convolution values are needed.

Graph:

- Shortest output graph.

Observation

1. The MP3 audio file was successfully loaded and converted into a mono signal.
2. The audio samples were extracted and converted into a NumPy array for processing.
3. A convolution operation was performed using the kernel **[1, 1, 1, 1, 1, 1, 1]**.
4. The **"full"** convolution mode generated the complete convolution output, making the output signal longer than the original.
5. The convolution process smoothed the audio waveform by reducing rapid fluctuations.
6. The amplitude of the convoluted signal was normalized to avoid clipping and distortion.
7. The processed signal was converted back to audio format and saved as **output_convoluted.wav**.
8. The original and convoluted waveforms were displayed using Matplotlib for comparison.
9. The convoluted waveform appeared smoother than the original waveform.

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10. The kernel acted as a simple moving average filter, reducing high-frequency components in the audio signal.
11. Edge effects were visible at the beginning and end of the convoluted signal because the "**full**" mode was used.
12. The experiment confirmed that convolution changes the characteristics of an audio signal based on the selected kernel.

Conclusion

1. The program demonstrates the convolution of an audio signal using Python.
2. The input MP3 file is successfully converted into a mono audio signal.
3. The audio signal is represented as numerical samples for processing.
4. Convolution is performed using the kernel **[1, 1, 1, 1, 1, 1, 1]**.
5. The selected kernel acts as a moving average filter.
6. The convolution operation smooths the audio waveform by averaging neighboring samples.
7. High-frequency components and sudden variations in the signal are reduced after convolution.
8. The convoluted signal is normalized to prevent clipping and maintain proper amplitude.
9. The processed signal is converted into a WAV file and saved as **output_convoluted.wav**.

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10. The program plots both the original and convoluted audio signals for comparison.
11. The convoluted waveform appears smoother and less noisy than the original waveform.
12. The **"full"** convolution mode produces the complete convolution result.
13. The output signal becomes longer than the original because of the **"full"** mode.
14. Edge effects are present at the beginning and end of the convoluted signal.
15. The experiment shows how different kernels can modify the characteristics of an audio signal.
16. Different kernels can be used for smoothing, sharpening, or edge detection.
17. The program provides a practical understanding of the convolution operation in Digital Signal Processing (DSP).
18. Overall, the experiment demonstrates how convolution is used for audio filtering and signal analysis.